

Communication Technology Research

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

1. Communication architecture

Channel	Data	Safety dependency
ESP-NOW	Fast peer truck beacons	Helpful but not required for local radar safety
LoRa	Fog posts, emergency beacon, long-range relay	Not the primary safety loop
Local safety MCU	Radar + corridor + alerts	Independent of network
Ops dashboard	Analytics, closures, replay	Read-only with no path into truck safety loop

2. Message design

Each message carries age, sequence number and signature. ProofRoad uses age counters instead of a shared clock because a GPS-poor pit may not provide reliable common time (p. 9).

Messages are classified according to consequence. Caution-adding messages are accepted if valid; restriction-relaxing messages require stronger local verification (p. 10).

3. Threat model

Threat	Expected behavior
Replay	Reject using sequence/age state and log
Forged hazard/hold	May cause nuisance slowdown but cannot increase speed
Forged GO/clear	Needs device signature plus local check
Jamming	Silence becomes uncertainty; gate waits; peer data becomes stale
GNSS spoof/jam	Use route/IMU/wheel cross-check; growing error shortens corridor
Compromised dashboard	Cannot alter truck safety advice because dashboard is outside safety loop

4. Production security direction

ProofRoad proposes per-device keys in secure elements, revocation, signed firmware, secure boot and rollback protection for production. It references ISA/IEC 62443 practice but explicitly does not claim compliance (p. 10).

For a student prototype, document the demo-key limitation clearly and demonstrate replay rejection, forged-GO rejection and radio-loss behavior.

5. Communication test plan

Measure packet delivery ratio, latency, range, message age, duplicate/replay behavior, interference sensitivity and recovery time.

Test the exact fail-safe behavior with the network disconnected, jammed, delayed and restarted.